

EE3621: Digital Signal Processing

Lecture 18: IIR Filter - Cascade Form Realization (III B.Tech EEE)

September 23, 2026

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Factor** high-order IIR transfer functions into second-order biquad sections (SOS).
 2. **Draw** complete Cascade realization signal flow graphs.
 3. **Analyze** computational complexity of cascade structures.
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2. MATHEMATICAL FOUNDATIONS

2.1 The Cascade Realization Formulation

$$H(z) = g \prod_{k=1}^K H_k(z) = g \prod_{k=1}^K \frac{b_{k0} + b_{k1}z^{-1} + b_{k2}z^{-2}}{1 + a_{k1}z^{-1} + a_{k2}z^{-2}}, \quad K = \lceil N/2 \rceil$$

Each biquad $H_k(z)$ realizes one real pole pair and one real zero pair:

$$1 + a_{k1}z^{-1} + a_{k2}z^{-2} = (1 - p_k z^{-1})(1 - p_k^* z^{-1}) = 1 - 2r_k \cos(\theta_k)z^{-1} + r_k^2 z^{-2}$$

Pictorial Representations

Cascade (Serial) IIR realization (SOS 1 followed by SOS 2)



Figure 1: Iir Cascade

3. WORKED NUMERICAL EXAMPLES

Example 18.1: 4th-Order IIR Cascade Realization

Problem: Realize $H(z) = \frac{(1+z^{-1})(1-\sqrt{2}z^{-1}+z^{-2})}{(1-0.9e^{j\pi/4}z^{-1})(1-0.9e^{-j\pi/4}z^{-1})(1-0.6z^{-1})(1+0.5z^{-1})}$ in Cascade Form with optimal pairing.

Solution:

1. Poles:

* Complex pair: $p_{1,2} = 0.9e^{\pm j\pi/4} \implies 1 - 2(0.9)\cos(\pi/4)z^{-1} + 0.81z^{-2} = 1 - 1.2728z^{-1} + 0.81z^{-2}$. (Radius $r = 0.9$, high Q). * Real poles: $p_3 = 0.6$, $p_4 = -0.5 \implies (1 - 0.6z^{-1})(1 + 0.5z^{-1}) = 1 - 0.1z^{-1} - 0.3z^{-2}$. (Low Q).

1. Zeros:

* Complex pair: $z_{1,2} = 1 \cdot e^{\pm j\pi/4} \implies 1 - \sqrt{2}z^{-1} + z^{-2} = 1 - 1.4142z^{-1} + z^{-2}$ (at angle $\pi/4$). * Real zeros: $z_3 = -1 \implies (1 + z^{-1})$.

1. Cascade Sections:

* Section 1: $H_1(z) = \frac{1+z^{-1}}{1-0.1z^{-1}-0.3z^{-2}}$. * Section 2: $H_2(z) = \frac{1-1.4142z^{-1}+z^{-2}}{1-1.2728z^{-1}+0.81z^{-2}}$.

1. **Realization:** $H(z) = H_1(z)H_2(z)$.